SMARTAG OPS ADVISOR

Prompt Document

Every prompt that shaped the application: what was asked, what was built, and reusable versions

25 October 2026 · Repository hajmalirfan/Data-Drive

An analysis of every prompt used to build and change SmartAg Ops Advisor with Claude Code. It also gives a clear, reusable version of each prompt, with acceptance criteria. Use it to review how the app was built, to repeat or extend a feature, or to write better prompts for the next one.

Project	SmartAg Ops Advisor (`hajmalirfan/Data-Drive` , folder `agri-project/`)
Session covered	24 Oct 2026, 17:25 IST to 25 Oct 2026, 03:45 IST
Messages analysed	44 user messages: 30 requests, plus option picks, screenshots and pasted HTML
Earlier work	7–24 Oct 2026, known only from git history (section 5)
Companion document	Master Document (MASTER_DOCUMENT.pdf): what the app is and how it works

> **Secrets are left out on purpose.** Several prompts pasted live credentials (Gemini API key, Neon database URL with password, Mailtrap token, data.gov.in API key). They appear here only as `[REDACTED]` . Because they were pasted into a chat, treat them as exposed and **rotate all four** (see section 7).

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1. Summary of the analysis

By area

Area	Requests	Prompt numbers
Setup and environment	4	P1–P4
Accounts, admin and security	7	P5, P10–P13, P15, P26
Truck operator and profile	4	P7–P9, P26
Maps and live tracking	4	P6, P20, P21, P28
Market data and prices	3	P18–P20
AI assistant	2	P14, P22
Layout and dashboards	3	P16, P24, P13
Documents and reports	3	P17, P23, P30
Git (commit and push)	3	P25, P27, P29

Some requests touch more than one area, so the counts add up to more than 30.

How the prompts were written

- **Element plus instruction (most common).** The prompt pastes the HTML of a screen element (a nav link, a form, a map) and adds a short instruction such as "remove the it admin role only". The pasted HTML pinned down exactly which screen was meant, so these needed no follow-up questions.
- **Short commands.** For example "run the application frontend and backend and database", "commit on github and push", "FIX THE APPLICATION ALLIGMENT IN FULLY". These worked when the scope was clear. "Fix the alignment fully" was the broadest prompt; it covered the whole app and was checked page by page with desktop and phone screenshots.
- **Screenshots as follow-ups.** Phone and desktop screenshots were sent after a change to point at what still looked wrong.
- **Course corrections.** Prompts P18 to P19 changed direction: the data.gov.in API sync was interrupted and replaced with "use the dataset I uploaded".
- **Firm data rule.** P18 set a rule that shaped later work: "everything is original, not fake data, only retrieved data". Mandi prices, the market finder and the revenue chart all show only real records. Missing values are labelled, never estimated.

Outcome

All 30 requests were completed. The work from P5 onward is in git: the commits listed in section 2 were pushed to `main` on GitHub.

2. Prompt log

Times are IST. **Original** is your wording, lightly cleaned (spelling kept where it carries meaning, pasted HTML summarised in square brackets). **Commit** is where the change landed. Commits marked † were made in a separate session or by hand between these prompts, alongside this work.

#	Time	Original prompt (cleaned)	What was delivered	Commit
P1	24 Oct 17:25	"Fetch and follow instructions from … superpowers/…/INSTALL.md", then chose option 2	Superpowers plugin installed from its marketplace (the INSTALL.md link returned 404, so the README steps were used)	n/a
P2	17:27	"run the application frontend and backend and database"	Backend (port 8000), frontend (port 5173) and database started	n/a
P3	17:38	[Gemini API key, `[REDACTED]`]	Key stored in `backend/.env` only	n/a
P4	17:43	[Neon PostgreSQL URL, `[REDACTED]`] "remove the old local database add the new database"	App switched to Neon; connection made resilient (`pool_pre_ping`, `pool_recycle`)	n/a
P5	17:52	[Mailtrap sample code] "admin role login on normal login page … mail id and default password 123456 … OTP will be sent … not on farmer and truck operator"	Admin sign-in needs password **and** an emailed one-time code; farmers and truck operators sign in without OTP; admin can't be created through public sign-up	`46b6378`
P6	18:26	[trip map HTML] "map have small size, add the button to full screen … like a google map design"	Compact trip map with a Google Maps-style full-screen view, zoom, recentre and map/satellite switch	`24bd295`
P7	18:57	[profile form HTML] "add two input boxes: permit and driver licence upload … licence has two sides … image will store on database"	Permit, licence front and licence back image uploads, stored in the database (`user_documents`)	`24bd295`
P8	19:13	[profile and documents HTML] "the fields are merged in one box … save will save and reset will reset, the tab will close"	Uploads moved into the profile form; Save uploads and saves together; Reset discards and closes	`24bd295`
P9	19:16	[Edit profile heading] "the user can click edit to open … otherwise closed"	Edit profile panel opens only when clicked	`24bd295`
P10	19:31	[nav links] "list out the features, remove for the admin role"	My Crops, Find Markets and Truck Booking hidden and blocked for admins	`24bd295`
P11	19:34	[nav links] "remove it for the admin role only"	Market Prices, Live Tracking and My Bookings hidden and blocked for admins	`24bd295`
P12	19:36	[admin user table HTML] "admin can remove the user, remove in database also, add the button"	Remove button with confirmation; account deleted from the database; guards against removing yourself or the last admin	`24bd295`
P13	19:45	[dashboard HTML] "admin dashboard: how many users, farmer, truck counts, AI chatbox usage graph, application request monitor graph"	Admin dashboard with user counts by role, AI chat usage chart and API request chart (24 h / 7 d / 30 d)	`24bd295`
P14	19:57	[chat textarea HTML] "add the mic voice recognition, multiple language support with proper backend"	Mic button; speech sent to the backend and transcribed by Gemini in the selected language	`24bd295`

#	Time	Original prompt (cleaned)	What was delivered	Commit
P15	21:49	"add the application security policy in full application, whatever is mandatory add the policy"	`SECURITY.md` ; password policy, login lockout, rate limits, security headers, CSP, strict CORS, session revocation, audit log	`8271469`
P16	22:05	"FIX THE APPLICATION ALLIGMENT IN FULLY" (+ phone screenshots)	Layout fixes across pages for desktop and phone	`8271469`
P17	23:01	"give me a master document for my application"	`docs/MASTER_DOCUMENT.md`	`f68f432`
P18	23:19	"data.gov.in mandi resource … using the [API key `[REDACTED]`] every day updated … everything is original, not fake data, only retrieved data"	API sync built; test interrupted by you	(became P19)
P19	23:35	"i upload the data in dataset, use it"	Government mandi prices loaded from `dataset/tamil_nadu_tomato_mandi.csv` (173 rows); file re-checked every 5 minutes; API mode kept as an option	`e61498e`
P20	23:51	[Find Markets nav] "backend calculates the farmer's nearest best price … transport charge and labour … Nearest, Quickest trip, Best crop price … 100% profit for farmer … Leaflet map with proper backend"	Market finder: real road distance and time, transport and labour costs, net in hand; picks for Nearest, Quickest, Best price, Best profit; Leaflet map	`e61498e`
P21	25 Oct 00:24	[Leaflet map HTML] "add the full screen button like google map"	Full-screen market map with route button and map/satellite switch	`e61498e` / `66cbea3`
P22	00:41	"ai assistant chatbox is not working properly, chat will loading"	Cause: the main Gemini model was overloaded. Fix: falls back to a lighter model after 20 s; the browser gives up after 75 s of silence; "Still thinking…" note	`66cbea3`
P23	01:00	"my full prompt convert to generate the pdf report"	`docs/SmartAg_Work_Report.pdf` (10 pages)	†
—	01:19–03:02	(work outside this chat)	Hosted backend on Render, CORS for the deployed frontend, mobile layout, real dashboard data and map, first revenue chart	`881bcfb` – `a260d50` †
P24	03:06	[empty "No prices recorded" card] "remove the feature, add another feature: revenue graph"	Price card removed from the farmer dashboard; revenue chart moved into its place	`e046303`
P25	03:09	"commit on github and push"	Pushed	`e046303`
P26	03:12	"truck operator role, set the * mandatory input boxes, set the policy"	Red * on mandatory fields in the operator profile and Add truck forms; enforced in the browser **and** the backend; saved permit and licence images can be replaced but not removed	`ade66c4`
P27	03:17	"commit on github and push it"	Pushed	`ade66c4`

#	Time	Original prompt (cleaned)	What was delivered	Commit
P28	03:24	"truck driver can share the location each 15 sec to the farmer … truck can move, farmer map will show changing coordinates every 15 sec … not a single coordinate to stop"	GPS sent every 15 s from anywhere in the app; resumes after reload; route trail stored; farmer map shows the moving truck, its path and "updated N s ago"	`3b0677d`
P29	03:35	"commit on github and push it"	Pushed	`3b0677d`
P30	03:45	"analysis my full project prompts to create the prompt document and also create master document"	This document and the updated master document	(this change)

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3. Refined, reusable prompts

Each prompt below says the same thing as the original but is complete on its own: it names the screen, the roles, the rules, and how to check the result. Reuse them to rebuild a feature, port it, or hand it to another developer.

Setup

P2 · Run the app

> Start the SmartAg backend (FastAPI, `agri-project/backend`, port 8000) and frontend (Vite, `agri-project/frontend`, port 5173) against the database in `backend/.env`. Confirm `GET /api/health` returns 200 and the login page loads.

P3–P4 · Configure services

> Put the Gemini API key and the Neon PostgreSQL URL in `backend/.env` only, never in code, docs or git. Remove the old local database setting. Make the database connection survive Neon idle disconnects. Confirm the app starts and can sign in.

Accounts, admin and security

P5 · Admin sign-in with email OTP

> Admins sign in on the normal login page with email and password, then must enter a 6-digit code emailed through Mailtrap. Farmers and truck operators sign in without a code. Codes expire after 10 minutes, allow 5 attempts, and can be resent after 60 seconds. The intermediate login token must not work as a session. Public sign-up must not be able to create an admin. Keep the Mailtrap token in `.env`.

P10–P11 · Hide farmer features from admins

> For the admin role, remove My Crops, Find Markets, Truck Booking, Market Prices, Live Tracking and My Bookings from the navigation, and redirect admins who open those URLs directly to the dashboard. Do not change other roles.

P12 · Remove a user

> On the admin user table, add a Remove button with an in-page confirmation. It deletes the account from the database. Block removing yourself and the last admin. A removed truck operator's trucks become inactive.

P13 · Admin dashboard

> Replace the admin's dashboard with: total users and counts per role (farmers, truck operators, others); a chart of AI assistant usage; and a chart of API requests, each switchable between 24 hours, 7 days and 30 days. Use stored, real counts only.

P15 · Security policy

> Write `SECURITY.md` and implement the mandatory controls: password rules, account lockout after repeated failures, rate limits on sign-in, OTP and AI endpoints, security headers and a Content Security Policy, CORS limited to known frontends, sign-out of old sessions after a password change, upload size limits, and an audit log of security events. Production must refuse weak secrets.

P26 · Mandatory fields for truck operators

> For the truck operator role, mark mandatory fields with a red * and enforce them in the browser and the backend. Profile: full name, mobile, village/town, vehicle type, vehicle number (3+ characters), vehicle capacity (> 0), permit, licence front and licence back. Add truck: vehicle number, truck type, status, base fare (≥ 0), rate per km (> 0), current location. Don't fill blank fields with defaults. Saved documents can be replaced but not removed. Farmers are not affected.

Truck operator profile

P7–P9 · Documents inside the profile form

> In the truck operator's Edit profile form, add image uploads for the vehicle permit and the driving licence front and back (JPEG, PNG or WebP, up to 5 MB, checked by file content). Store the images in the database. Picked images are previewed but saved only when Save is pressed; Reset discards changes and closes the panel. The Edit profile panel is closed until clicked. Admins can view a user's documents.

Maps and tracking

P6 / P21 · Map with full-screen view

> Show the map at a compact size with a full-screen button in the top-right corner, styled like Google Maps: full screen fills the window, Esc or the close button exits, and full screen adds zoom, recentre or route, and a map/satellite switch. Refit the view after entering or leaving full screen.

P28 · Live truck location every 15 seconds

> While a trip is picked up or in transit and the operator has given consent, the operator's phone sends its GPS position every 15 seconds, even if the position hasn't changed and while the operator uses other pages. Sharing resumes after a page reload and stops only when the operator presses Stop, the trip is delivered, or the server refuses the update. Store each position. The farmer's tracking map shows the truck moving, its route so far and "updated N s ago", and warns when no update has arrived for 45 seconds. Stopping or delivering removes the truck and its route from the farmer's view. Show a sharing bar with a Stop button on every page.

Market data

P18–P19 · Government mandi prices

> Load government mandi prices from the CSV files in `dataset/` (columns state, district, market, commodity, variety, grade, arrival_date, min/max/modal price). Keep prices exactly as published, in rupees per quintal; never estimate or invent values. Re-import when a file changes. Keep the data.gov.in API as an alternative source with the key in `.env`. Show source, date and unit with every price.

P20 · Find the best market

> For a chosen crop and quantity, rank markets from the farmer's location using the latest government modal prices, real road distance and travel time, the cheapest available truck's fare, and labour costs. Show net money in hand per market and pick out Nearest, Quickest trip, Best price and Best profit. Plot the markets on a Leaflet map with the route to the selected one. Leave out markets whose location or route can't be found, and say so.

AI assistant

P14 · Voice input

> Add a mic button to the AI assistant. Record in the browser, send the audio to the backend, transcribe it with Gemini in the selected app language, and put the text in the message box for the user to check before sending. Rate-limit the endpoint.

P22 · Chat stuck on loading

> The assistant sometimes loads forever. Find the cause. Make the backend fall back to a lighter model if the main one hasn't started answering within 20 seconds, and make the browser stop waiting after 75 seconds of silence with a clear error. Show "Still thinking…" after 12 seconds.

Dashboards and layout

P16 · Alignment

> Check every page at desktop (1400 px) and phone (390 px) widths. Fix misaligned, stretched, squashed or overflowing elements. Report each fix with before/after screenshots.

P24 · Farmer dashboard revenue

> On the farmer dashboard, remove the "Latest published market prices" card and put the revenue chart in its place: sales recorded after delivery over the last 6 months, with the total. Show an empty state instead of an empty chart.

Documents

P17 / P30 · Master and prompt documents

> Write a master document for the application (scope, roles, architecture, features, data model, API, configuration, setup, security, constraints) and a prompt document that lists every prompt with what was delivered, a reusable version of each, and lessons. Leave out all secrets. Provide Markdown and PDF.

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4. One consolidated prompt for the whole application

Use this to rebuild the app or brief a new developer. It combines the feature requests above with the earlier work in section 5.

> Build **SmartAg Ops Advisor**, a web app that helps Indian farmers sell produce at the best mandi and move it by truck.
>
> **Stack:** React 18 + Vite + plain CSS frontend; FastAPI + SQLAlchemy 2 + Pydantic 2 backend; PostgreSQL (Neon). Leaflet maps. Ten UI languages: English, Hindi, Tamil, Telugu, Kannada, Malayalam, Marathi, Bengali, Gujarati, Punjabi. All secrets in `backend/.env`, never in code or git.
>
> **Roles:** farmer and truck operator sign up publicly; admin is provisioned from configuration. Admin sign-in needs password plus emailed OTP (Mailtrap). The backend enforces every role and ownership rule.
>
> **Farmer:** crop stock; government mandi prices (from the dataset CSV, or the data.gov.in API), shown as published with source, date and unit; a market finder ranking markets by net money in hand after real road transport and labour costs (Nearest, Quickest, Best price, Best profit) on a Leaflet map; truck booking; live trip tracking with the truck moving every 15 seconds; a dashboard with crop stock, nearby markets and a 6-month revenue chart of recorded sales; reports; an agriculture AI assistant (Gemini with fallback model, streaming, voice input).
>
> **Truck operator:** profile with mandatory vehicle details, permit and both licence images (stored in the database); trucks with mandatory type, fares and location; booking requests; trips; live GPS sharing every 15 seconds with consent, a sharing bar on every page and a Stop button.
>
> **Admin:** dashboard with user counts by role, AI usage and API request charts; user list with role change and removal (not yourself, not the last admin); document review; mandi sync status. Farmer-only pages are hidden from admins.
>
> **Rules:** only real data; never invent prices, locations or routes; label missing values. Security per `SECURITY.md` (password policy, lockout, rate limits, headers, CSP, strict CORS, session revocation, audit log). Every page must work at 1400 px and 390 px widths.

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5. Work before this session (from git history)

The prompts for this period aren't in the transcript, so only the commit messages are known. They are listed so the project history is complete.

Date	Commit	Change
7 Oct	`d69a719`	Initial commit
8 Oct	`f7542a7`	FastAPI auth (sign-up, login, JWT), React login, layout, 10-language i18n
12 Oct	`aa60378`	Roles (farmer, FPO, marketer, admin) with access control
21 Oct	`1bc8511`, `3529a17`	Full multilingual support and the complete project merged
22 Oct	`a91e1b4`	PostgreSQL; demo seed data removed
23 Oct	`5483ecb`	New login page; FPO role removed
23 Oct	`88f316f`	Password reset with OTP through Mailtrap
23 Oct	`64f627c`, `c4aed15`	Truck operator fleet workflows; sign-up limited to farmers and truck operators
23 Oct	`3bd0121`, `5441b7d`	Gemini assistant; live trip tracking; booking access fix
24 Oct	`b5aa2dc`	Gemini/Ollama switch and booking workspace

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6. What worked well, and what to do differently

Worked well

- Pasting the element's HTML made the target unmistakable: no prompt that did this needed clarification.
- Stating a firm rule ("original data only") once, early, kept every later feature honest without repeating it.
- Interrupting quickly (P18) when the direction was wrong saved time.
- Asking for "commit and push" after each feature kept GitHub in step with the work, one feature per commit.

Do differently

- **Don't paste secrets into the chat.** Put them straight into `backend/.env` and say "the key is in .env". Pasted secrets end up in transcripts.
- **Say who the change is for and how you'll check it.** For example "for truck operators only; done when the farmer's map moves every 15 s". P26 and P28 were clear about the result; P16 ("fix alignment fully") had no end point.
- **One feature per prompt.** P13 bundled counts, two charts and a removal; it worked, but smaller prompts are easier to review and roll back.
- **Mention where the app runs.** The frontend talks to the Render backend, so backend changes reach the app only after a push and redeploy. Saying "test on Render" or "test locally" up front avoids confusion.
- **Name the language scope.** New messages were added in English only; ask for translations in the same prompt if they're needed.

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7. Open items raised by the prompts

Item	Why	Suggested action
Rotate exposed secrets	Gemini key, Neon password, Mailtrap token and data.gov.in key were pasted in chat	Create new ones, update `backend/.env` and Render, revoke the old ones
Default admin password	P5 asked for `123456`, which fails the password policy from P15	Change it after first sign-in; the app logs a warning until then

Item	Why	Suggested action
Translations	Messages added in P26 and P28 are English only	Add them to the language catalogue and regenerate
Dataset coverage	The dataset is Tamil Nadu tomato only (173 rows, 24 Oct 2026)	Add more CSVs or switch `MANDI_SOURCE=api`
Background GPS	A website can't send GPS when the app is closed or the phone kills the tab	A native app would be needed for that
Automated tests	Changes were checked with throwaway databases and browser scripts, not a test suite	Add pytest and browser tests for sign-in, documents, finder and live tracking
Build brief conflict	`CLAUDE.md` in the Downloads folder describes a different, guest-only demo app	Decide which brief applies and remove or update the other

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8. Prompt template

Copy this for the next feature.

```
Feature: <one sentence>
For: <roles affected; say "only" if others must not change>
Where: <page or element; paste its HTML if helpful>
Rules: <data, security or business rules>
Done when: <what you will see or test>
Also: <translations? commit and push? test locally or on Render?>
```